

Geometry Test I

Name
Date
Course

Instructions

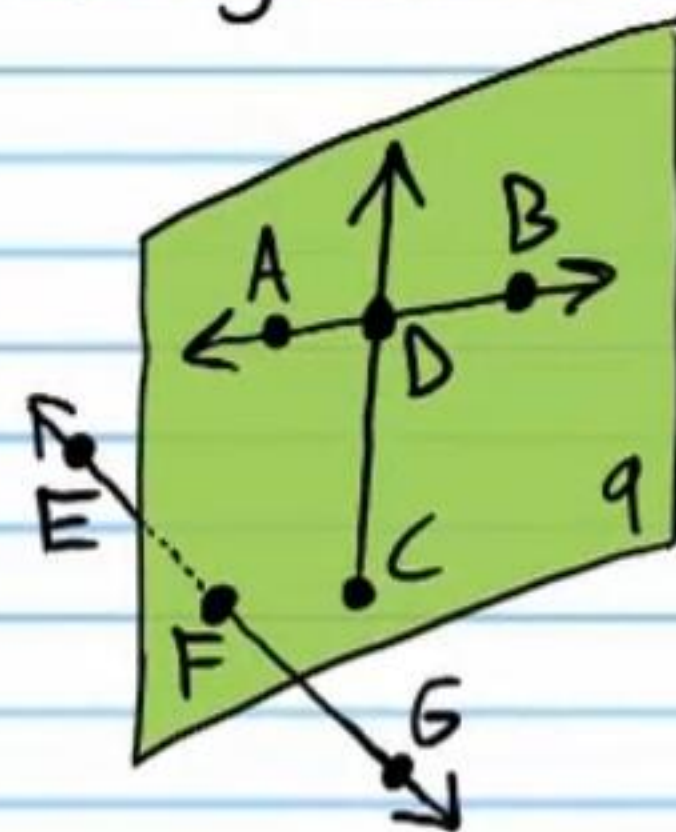
- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show work when necessary.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Each problem is worth four points for a total of 100 points.

Use diagram #1 below to answer question #'s 1-2.

Name the type of object in the diagram that each symbol refers to (e.g. some possible answers: plane, ray).

① i) q ii) $\overleftrightarrow{AB}$ Diagram #1

iii) $\overline{AB}$ iv) D



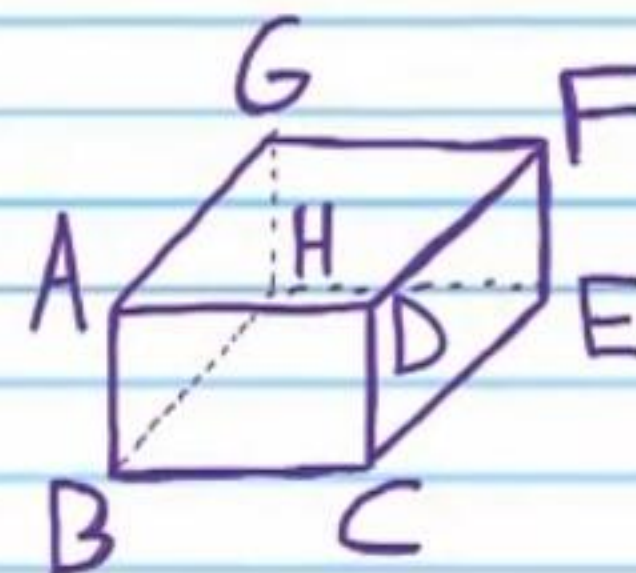
② i) What is another name for $\overleftrightarrow{AB}$? *There is more than one possible answer. Choose one.*

ii) Are E, F, and C collinear?

Use diagram #2 for problem #3.

③ i) Are A, G, and F coplanar? *Diagram #2*

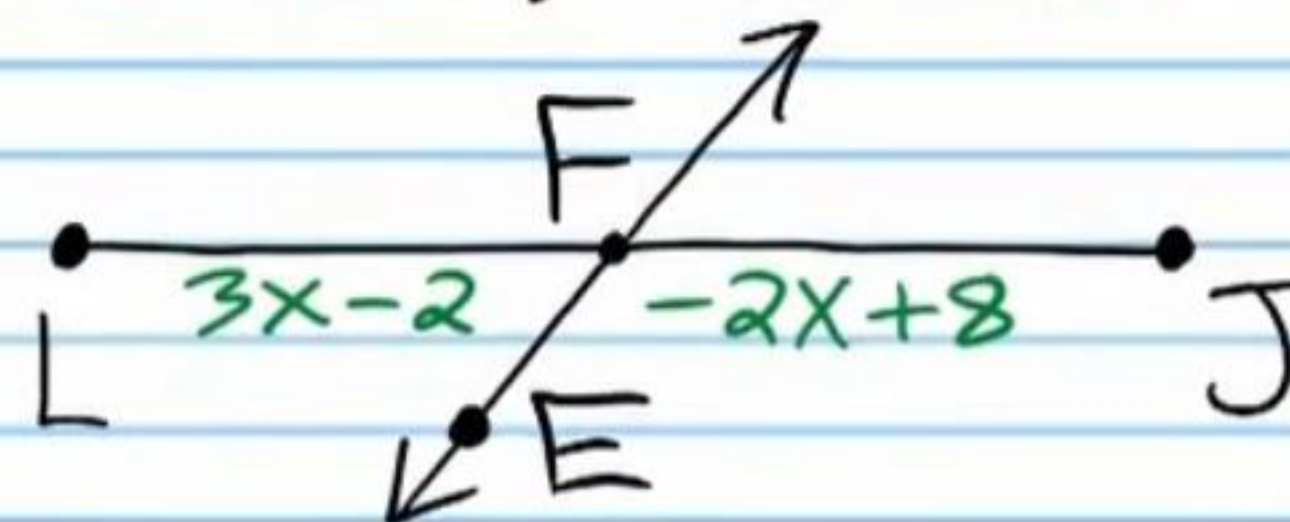
ii) Are B, D, and E collinear?



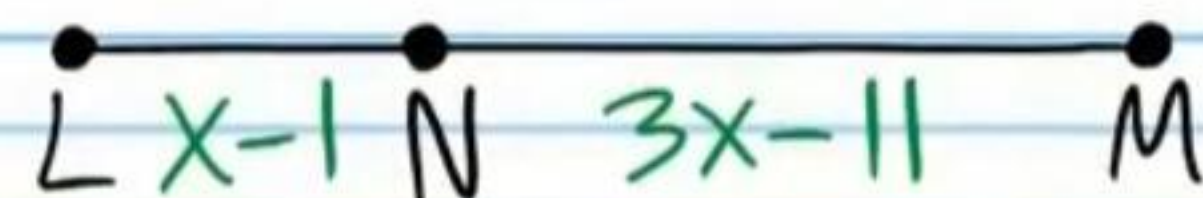
iii) Name all rays with endpoint B. *Hint: There are seven.*

iv) What is the intersection of $\overline{HE}$ and $\overline{CE}$?

④ If $\overleftrightarrow{EF}$ bisects $\overline{LJ}$ below, find FJ.



⑤ If $LM = 20$ below, find x .



⑥ Find the midpoint of the segment with endpoints $A(2, -6)$ and $B(2, -2)$.

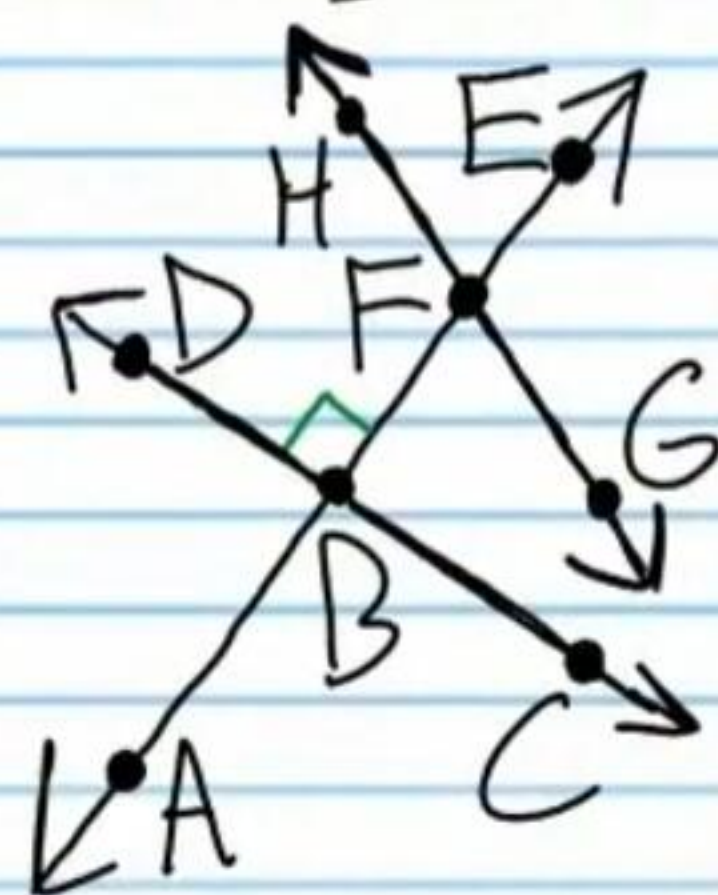
⑦ Classify the following angles (use diagram #3 for parts iii through iv).

i) 121°

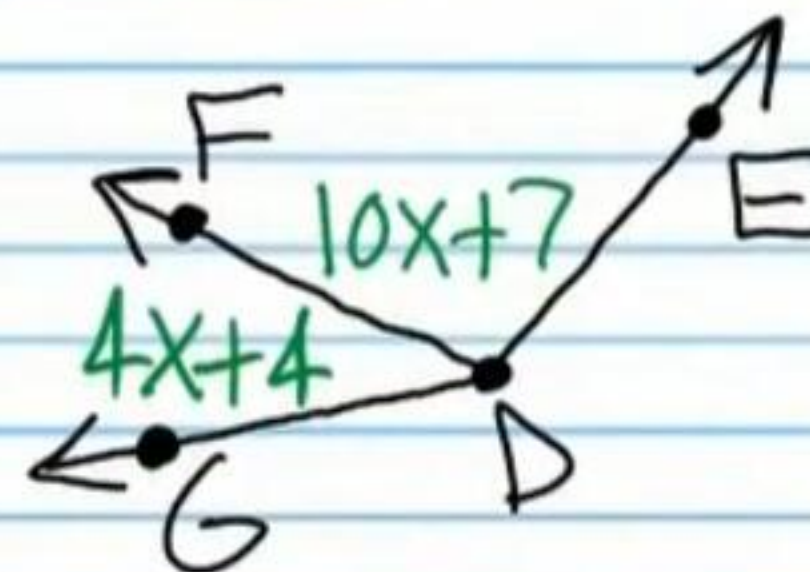
ii) 3°

iii) $\angle DBE$

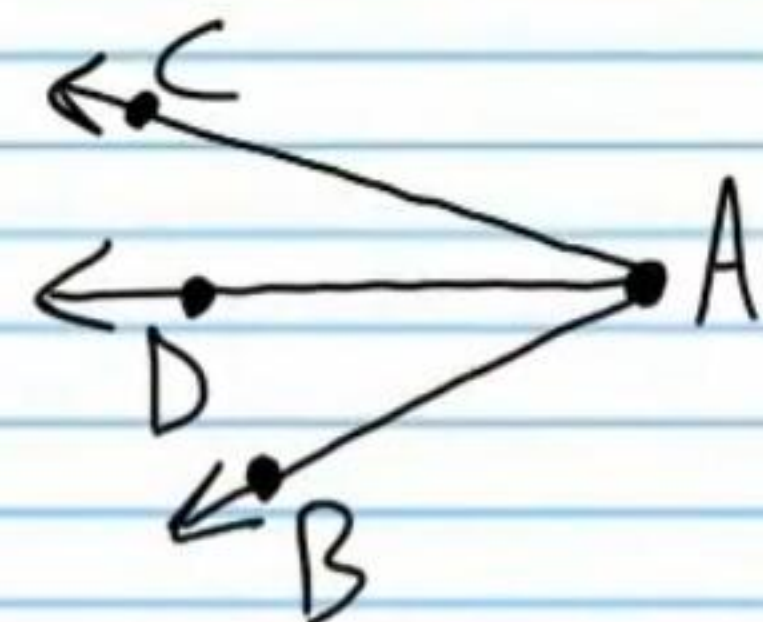
iv) $\angle DBC$



⑧ Find $m\angle FDE$ if $m\angle GDE = 137^\circ$.



⑨ If $\vec{AD}$ bisects $\angle BAC$, find $m\angle DAC$ if $m\angle BAC = 62^\circ$.



⑩ Write the steps required to construct a perpendicular bisector of $\overline{AB}$ below.



⑪ Use diagram #4 below.

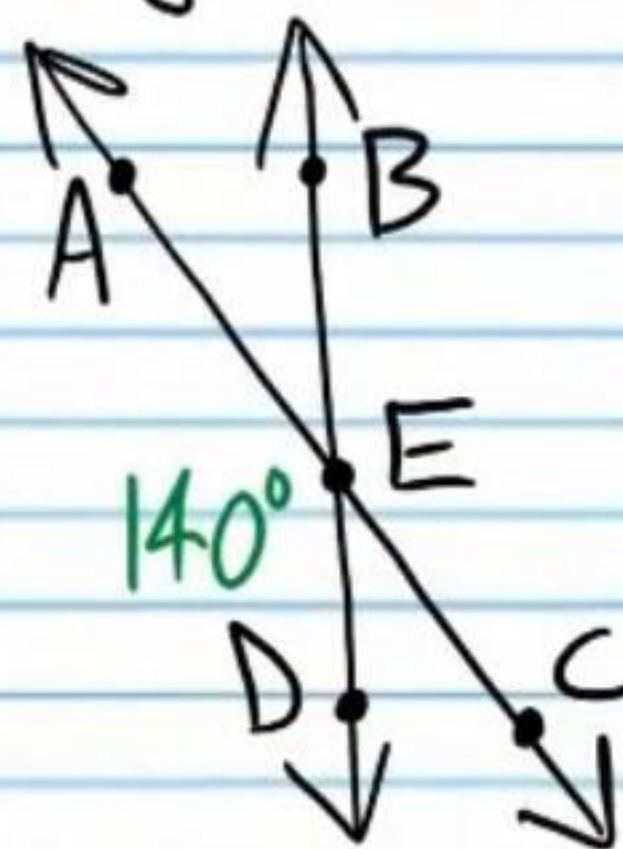
i) Name two angles in the diagram that are adjacent.

ii) Name two angles that are vertical.

iii) Name two angles that form a linear pair.

iv) Find $m\angle DEC$.

Diagram #4



⑫ Use diagram #5 below.

i) Name the polygon (i.e. classify it by the number of sides).

Diagram #5



ii) Is the polygon equiangular?

iii) Is it equilateral?

iv) Circle any figures below that are polygons.



⑬ Identify the following examples of reasoning as inductive or deductive.

i) Larry's first three phones were Apple brand. His fourth phone will also be Apple brand.

ii) If the two shorter sides of a right triangle are three inches and four inches, then the Pythagorean Theorem tells us that the longest side will be five inches.

iii) A heptagon is defined to be a seven sided polygon. Because a polygon shaped game board has seven sides, it is a heptagon.

iv) Andy used the Angle Addition Postulate to show that the measure of an angle is 23 degrees.

⑭ i) Make a conjecture for the next thing in the sequence.

$\Delta \rightarrow, \nabla \downarrow, \Delta \leftarrow, \dots$

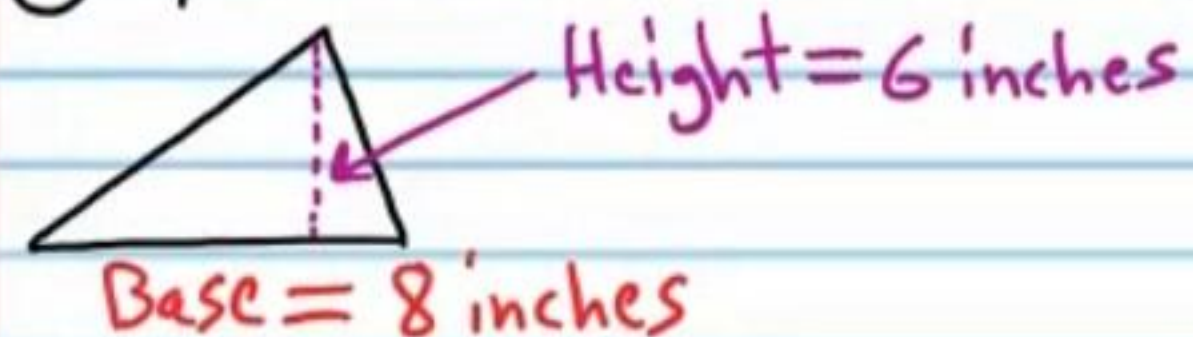
ii) Find a counterexample for the statement below.
Every dollar that I find is mine.

⑮ If I lose my wallet, then I will not be able to buy dinner.

i) Write the inverse of the statement above.

ii) Write the contrapositive of the statement above.

⑯ i) Find the area of the triangle below.



ii) Rewrite the following statement in if-then form.
Write the if-then form in a way that makes sense in context.

You must pay \$5 to enter the race.

⑰ Are the following statements biconditional? If so, rewrite in "if and only if" form.

i) If $3x+1=10$, then $x=3$.

ii) If Billy oversleeps, then he will be late to school.

⑱ Determine if the Law of Detachment or the Law of Syllogism can be used for the following examples. Make a valid conclusion using one of the laws.

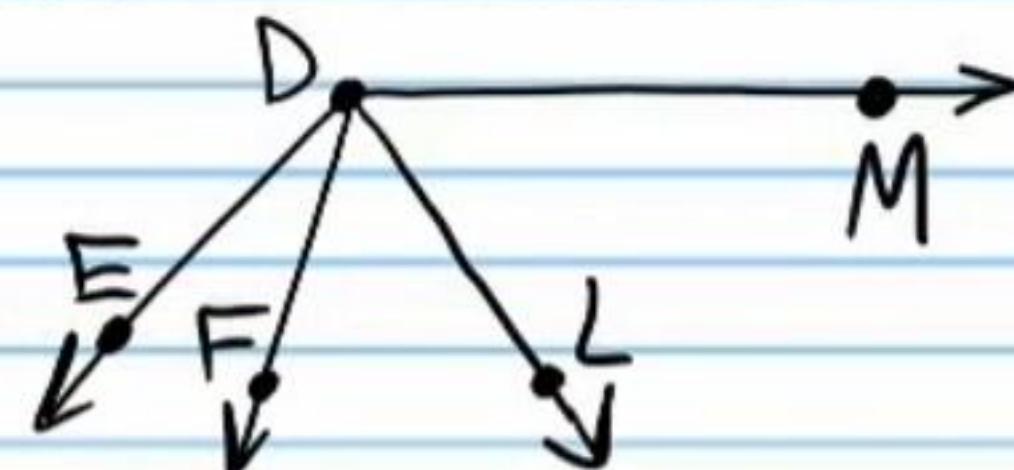
i) IF Arnold brings his backpack, then he will have food and water. IF he has food and water, then he will reach the summit before turning back.

ii) IF Kevin's football team wins today's game, then they will play against the Lions. IF they play against the Lions, then Kevin will get to see the City of Portland.

⑲ IF the measure of an angle is six degrees less than two times the measure of its complement, what is the measure of the angle?

Write a two-column proof for each statement below.

⑳ IF points "F" and "L" are in the interior of $\angle EDM$ as shown below (use this as the given information), then $m\angle EDM = m\angle EDF + m\angle FDL + m\angle LDM$.



Statement	Reason
I)	
II)	
III)	
IV)	

21) If the two segments below are congruent, then $x = -2$.

$$\overline{AB} = -8x + 6$$

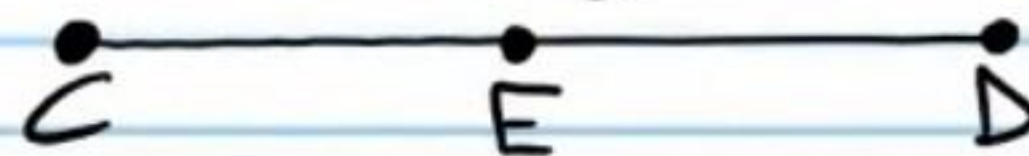
$$\overline{CD} = 12 - 5x$$

Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	
VIII)	

22) The converse of problem #21.

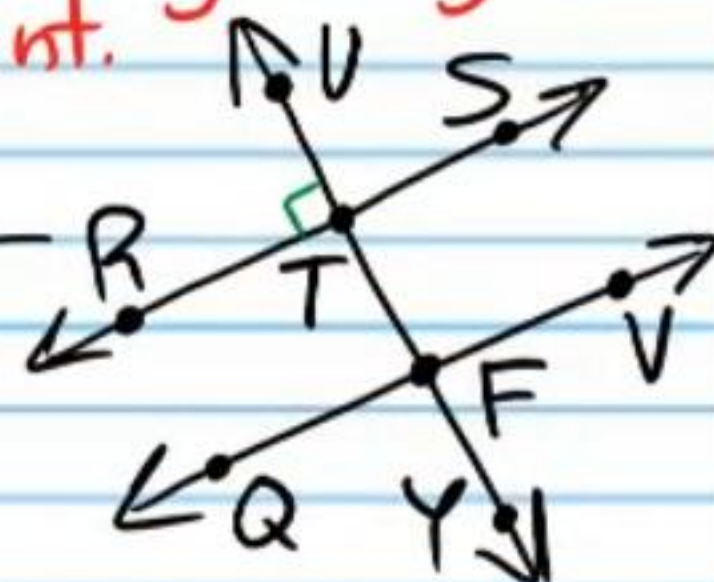
Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	

23) If E is the midpoint of the segment below, then $CD = 2CE$.



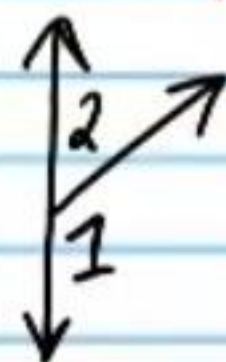
Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	

24) If $\overleftrightarrow{QU} \perp \overleftrightarrow{TY}$ and $\angle RTU$ is a right angle, then $\angle RTU \cong \angle TFW$. Use the Right Angle Congruence Theorem to prove this statement.



Statement	Reason
I)	
II)	
III)	
IV)	

25) If $m\angle 2 = 48^\circ$, then $m\angle 1 = 132^\circ$. Hint: You must use the Linear Pair Postulate.



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	

Geometry Test I Answers

① i) Plane

ii) Line

iii) Segment

iv) Point

② i) One of any of the following: $\overleftrightarrow{AD}$, $\overleftrightarrow{DB}$, $\overleftrightarrow{BD}$, $\overleftrightarrow{DA}$, or $\overleftrightarrow{BA}$.

ii) No

③ i) Yes. *Any three points are coplanar.*

ii) No.

iii) $\overrightarrow{BA}$, $\overrightarrow{BC}$, $\overrightarrow{BD}$, $\overrightarrow{BH}$, $\overrightarrow{BE}$, $\overrightarrow{BF}$, and $\overrightarrow{BG}$.

iv) E

④ $FJ = 4$

⑤ $x = 8$

⑥ $(2, -4)$

⑦ i) Obtuse

ii) Acute

iii) Right

iv) Straight

⑧ $m\angle FDE = 97^\circ$

⑨ $m\angle DAC = 31^\circ$

⑩ Put a compass on an endpoint of the segment and extend it beyond the midpoint of the segment. Make an arc intersecting the segment. Keeping the compass at the same extension, place it at the other end of the segment and draw another arc intersecting the first arc. Draw a line through the points where the arcs intersect.

⑪ i) Choose one of the following pairs: $\angle BEC$ & $\angle CED$ or $\angle AEB$ & $\angle BEC$ or $\angle AEB$ & $\angle AED$ or $\angle AED$ & $\angle DEC$.

ii) Choose one of the following pairs: $\angle BEC$ & $\angle AED$ or $\angle AEB$ & $\angle DEC$.

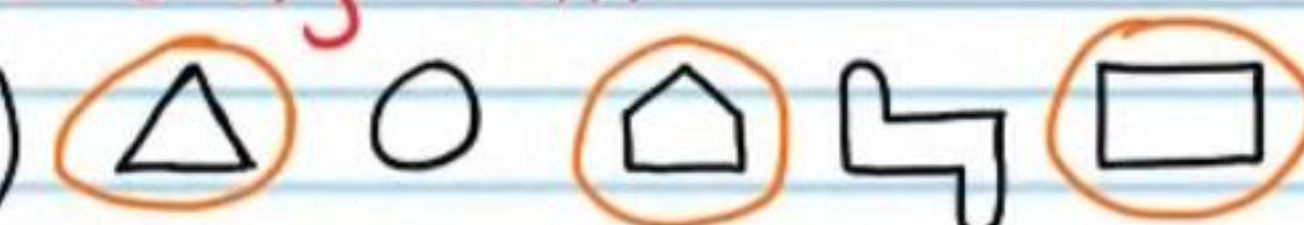
iii) Choose one of the following pairs: $\angle BEC$ & $\angle CED$ or $\angle AEB$ & $\angle BEC$ or $\angle AEB$ & $\angle AED$ or $\angle AED$ & $\angle DEC$.

iv) $m\angle DEC = 40^\circ$

⑫ i) Quadrilateral

ii) No. *You can see that some angles are much bigger than others.*

iii) Yes. *The red marks tell us that all sides are congruent.*

iv) 

(13) i) Inductive

ii) Deductive

iii) Deductive

iv) Deductive

(14) i) $\nabla \uparrow$

ii) A dollar that I find may belong to someone else (other answers might be acceptable).

(15) i) If I do not lose my wallet, then I will be able to buy dinner.

ii) If I will be able to buy dinner, then I will not lose my wallet.

(16) i) 24 in^2

ii) If you want to enter the race, then you must pay \$5. The converse of this would not make sense.

(17) i) Yes

$3x+1=10$ if and only if $x=3$.

ii) No

The order does not matter

(18) i) The Law of Syllogism

If Arnold brings his backpack, then he will reach the summit before turning back.

ii) The Law of Syllogism

If Kevin's football team wins today's game, then he will get to see the city of Portland.

(19) 58°

(20)

Statement	Reason
I) F and L are in the interior of $\angle EDM$.	Given.
II) $m\angle EDL = m\angle EDF + m\angle FDL$	$\angle$ Add Post.
III) $m\angle EDM = m\angle EDL + m\angle LDM$	$\angle$ Add Post.
IV) $m\angle EDM = m\angle EDF + m\angle FDL + m\angle LDM$	Subs. Prop of =.

(21)

Statement	Reason
I) $\overline{AB} \cong \overline{CD}$	Given
II) $AB = CD$	Def. of $\cong$ Segs.
III) $AB = -8x + 6$ & $CD = 12 - 5x$	Given (diagram)
IV) $-8x + 6 = 12 - 5x$	Subs. Prop of =.
V) $6 = 12 + 3x$	Add. Prop of =.
VI) $-6 = 3x$	Subt. Prop of =.

VII)	$-2 = x$	Div. Prop of $=$.
VIII)	$x = -2$	Sym. Prop of $=$.

(22)	Statement	Reason
I)	$x = -2$	Given.
II)	$AB = -8x + 6$ & $CD = 12 - 5x$	Given (diagram)
III)	$AB = -8(-2) + 6$ & $CD = 12 - 5(-2)$	Subs. Prop of $=$.
IV)	$AB = 22$ & $CD = 22$.	Simplification.
V)	$AB = CD$	Trans. Prop of $=$.
VI)	$\overline{AB} \cong \overline{CD}$	Def. of $\cong$ Segs.

(23)	Statement	Reason
I)	E is the midpoint of $\overline{CD}$.	Given.
II)	$\overline{CE} = \overline{ED}$	Def. of Midpoint
III)	$CE = ED$	Def. of $\cong$ Segs.
IV)	$CD = CE + ED$	Seg. Add Post.
V)	$CD = CE + CE$	Subs. Prop of $=$.
VI)	$CD = 2CE$	Simplification.

(24)	Statement	Reason
I)	$\overleftrightarrow{QV} \perp \overleftrightarrow{TY}$	Given.
II)	$\angle TFE$ is a right angle.	Def of $\perp$
III)	$\angle RTU$ is a right angle.	Given.
IV)	$\angle RTU \cong \angle TFE$	Rt. $\angle \cong$ Thm.

(25)	Statement	Reason
I)	$\angle 1$ & $\angle 2$ form a linear pair.	Given (diagram)
II)	$\angle 1$ & $\angle 2$ are supplementary.	Linear Pair Post.
III)	$m\angle 1 + m\angle 2 = 180$	Def. of Supp $\angle$'s
IV)	$m\angle 2 = 48^\circ$	Given.
V)	$m\angle 1 + 48 = 180$	Subs. Prop of $=$.
VI)	$m\angle 1 = 132^\circ$	Subt. Prop of $=$.